

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence
Topic: PEAS Framework and Task Environment Analysis
System: Agricultural Monitoring Drone for Crop Disease Detection

Task 1: PEAS Specification

Performance Measure:

1. Disease detection accuracy (target: greater than 95%)
2. Percentage of crop area covered per flight
3. Early detection rate — disease caught before spreading
4. False positive/negative rate in disease identification
5. Battery efficiency (area covered per charge)

Environment:

Agricultural farmland with varying terrain, weather conditions, and crop types. Key external factors include wind speed, sunlight intensity, humidity levels, rainfall, and crop density.

Actuators:

- Propulsion motors for directional flight control
- Gimbal-stabilized camera mount for precise imaging
- GPS-guided navigation system
- Wireless data transmission module
- Automated spray nozzles for targeted treatment

Sensors:

- High-resolution RGB cameras for visual crop inspection
- Infrared (IR) sensors for heat-based disease detection
- Multispectral imaging sensors for chlorophyll analysis
- GPS and IMU (Inertial Measurement Unit) for positioning
- Wind and humidity sensors for environmental monitoring

Task 2: Environment Classification

Dimension	Classification	Justification
Observable	Partially Observable	Drone cannot see underground root diseases or hidden undersides of leaves.
Agent	Single-agent	One drone operates independently without coordinating with other agents.
Deterministic	Stochastic	Wind, weather, and disease spread are unpredictable and random.
Episodic/Sequential	Sequential	Earlier detections affect future flight paths and treatment decisions.
Static/Dynamic	Dynamic	Crops grow and weather changes while the drone is operating.
Discrete/Continuous	Continuous	Flight path, sensor readings, and environmental data are all continuous.

Task 3: Critical Analysis

Question 1: Greatest Technical Challenge

The most challenging dimension is the Stochastic environment. Wind gusts, sudden weather changes, and unpredictable disease spread patterns make it extremely difficult for an AI engineer to design a reliable system. The drone must constantly adapt its flight path and decision-making in real-time despite these uncertainties, requiring advanced probabilistic models and robust control algorithms.

Question 2: Utility Function

A suitable utility function would balance Coverage Speed vs. Detection Accuracy:

$$U = (\alpha \times \text{Detection Accuracy}) + (\beta \times \text{Area Coverage Rate})$$

Where alpha and beta are weights. If $\alpha = 0.6$ and $\beta = 0.4$, the drone prioritizes accuracy. If alpha's weight is doubled ($\alpha = 1.2$), the drone would slow down significantly to scan every plant more carefully, reducing area covered per flight but catching diseases much earlier — potentially missing large-scale outbreaks due to slower overall coverage.